

Carleton University
Department of Systems and Computer Engineering
SYSC 3303A RealTime Concurrent Systems Winter 2025
Assignment 5 - Rate Monotonic Scheduling

1. Consider three processes P1, P2, and P3 that have the following characteristics:

Process	Period	Execution Time
P1	8	3
P2	12	4
P3	20	3

- Draw timing diagram to show how these processes can be scheduled using the rate monotonic scheduling algorithm.

2. Consider three processes P4, P5, and P6 that have the following characteristics:

Process	Period	Execution Time
P4	100	15
P5	5	1
P6	25	10

- a) What is the processor utilization of P4, P5, and P6?
- b) If the processes were scheduled using the rate monotonic scheduling algorithm, would they meet their deadlines? Explain your answer.
- c) Draw timing diagram to show how these processes can be scheduled using the rate monotonic scheduling algorithm.

3. Consider three processes P7, P8, and P9 that have the following characteristics:

Process	Period	Execution Time
P7	70	30
P8	40	10
P9	20	5

- If you want to check the schedulability of these processes using the rate monotonic scheduling algorithm, you find out that they do meet all deadlines despite that they won't pass the Liu/Layland utilization test.
- Draw timing diagram to show how these processes can be scheduled using the rate monotonic scheduling algorithm.

4. Consider the following tasks (T1, T2, T3, T4, and Ta). The tasks T1-T4 are scheduled using rate monotonic algorithm and task Ta is has highest priority and not scheduled using RMS.

■ Using the Generalized Utilization Bound Theorem will each of these 5 tasks meet the following four factors that needs to be fulfilled?

1. Preemption by higher priority tasks with **periods less than** t_i . Tasks in this set can preempt t_i many times.
2. Execution time for task t_i . $U = C_i/T_i$
3. Preemption by higher priority tasks **with longer periods**. Can only preempt t_i once.
 $t_i = C_j/T_i$
4. The **maximum** blocking time B_i by lower-priority tasks holding a resource $B_i = B_j/T_i$

Task	Period	Execution Time
T1	108	12
T2	125	30
T3	175	25
Ta	200	10
T4	290	35

Work Products

1. A file or files containing timing diagrams for questions 1, 2, and 3. Hand drawn scanned diagrams are acceptable, as long as they are neatly drawn and your handwriting is legible, and the software required to view them is present in the lab.
2. Timing diagrams must be implemented in a similar format as the examples shown in the lectures. You must show the state of each of the tasks (idle, preempted, running, ...)

Reminder

The TAs will mark your assignments in the lab environment. It is your responsibility to ensure that any software required for viewing your files is also present in the lab.

Submitting Assignments

1. Submit the timing diagrams produced on Brightspace. **Missing any of the diagrams will result in receiving a zero on the assignment even if you got 100% on the quiz.**
2. This assignment is to be completed by using the quiz on Brightspace. For all questions, the values entered should be with a 4 decimals, e.g. 0.1 should be entered 0.1000.